Prefix/Infix/Postfix – Senior Division

1. **00-01 C4 Prefix/Infix/Postfix**

   Translate the following postfix equation to its equivalent prefix equation:

   a 2 ↑ b 2 ↑- a b + a b - * =

2. **00-01 C4 Prefix/Infix/Postfix**

   For this problem, we define 2 new binary operators.
   a # b = average of a and b
   a % b = harmonic mean of a and b (i.e. square root of a*b)
   Evaluate the following prefix expression:
   % + # 1 5 # 2 8 2

3. **01-02 C4 Prefix/Infix/Postfix**

   If A = 7, B = 9, C = 3, D = 4, E = 6, and F =8, evaluate the following postfix expression:

   A B C D E A - + ***** / + F E - +

4. **01-02 C4 Prefix/Infix/Postfix**

   Find all integer values of X for which the following prefix expression has a value of zero:

   ***** + X 4 - 6 X

5. **02-03 C4 Prefix/Infix/Postfix**

   Suppose **@** is a ternary operator whose value is the minimum of its three operands.
   Evaluate the following postfix expression:

   7 3 + 2 6 * 2 3 ↑ 3 + @

6. **02-03 C4 Prefix/Infix/Postfix**

   Convert the following prefix expression into an equivalent postfix expression keeping the order of operations the same:

   / - + ↑ X 2 * * 2 X Y ↑ Y 2 – X Y

7. 03-04 C4 Prefix/Infix/Postfix

Translate the following infix expression to postfix:

$$\frac{(A^2 + B)^2 - (AB + B^2)^{\frac{1}{2}}}{A/B^2 + A^2B/C}$$

8. 03-04 C4 Prefix/Infix/Postfix

The new binary operations **@** and # are defined as follows:

A @ B = $\sqrt{A^2 + B^2}$ and A # B = $\frac{A+B}{2}$

Evaluate the following prefix expression. Note that all numbers are single digit.

@ @ # 2 4 # 4 4 * 4 # 1 5

9. 04-05 C4 Prefix/Infix/Postfix

Define A # B = $\sqrt{AB}$. Evaluate the following prefix expression.
Note that all numbers are single digit.

* + * # 3 4 # 6 2 * ↑ 2 3 3 / * 4 5 + 2 3 1

10. 04-05 C4 Prefix/Infix/Postfix

Convert the following postfix expression to prefix

A B C + / B 2 ↑ C B / + - 2 ↑

11. 05-06 C4 Prefix/Infix/Postfix

Given $A = 3, B = 4, C = 8$ and $D = 2$ evaluate the following postfix expression.
Note that all numbers are single digit.

AB * C D - / 2 ↑ B 3 ↑ C 2 ↑ / + C D 2 ↑ * B 1 2 / ↑ / -

12. 05-06 C4 Prefix/Infix/Postfix

Keeping the order of operations the same, convert the following prefix expression to postfix:

↑ - + / A B / + A B – A B / * A B / A B 2

13. 06-07 C4 Prefix/Infix/Postfix

Convert the following infix expression to prefix:

$$\frac{\sqrt{\frac{A^2+(B-C)^2}{AB-C^2}}}{\left[\frac{(A-BC)^4}{B-C}\right]^{\frac{1}{3}}}$$

14. 06-07 C4 Prefix/Infix/Postfix

Define A # B = A^2 + B. Evaluate the following postfix expression given A = 2 and B = -1.

A B # 2 ↑ B A # A A B # # * -

15. 07-08 C4 Prefix/Infix/Postfix

Convert the following prefix expression to postfix:

/ * + A B – B C ↑ A C

16. 07-08 C4 Prefix/Infix/Postfix

Define A @ B = gcf (A, B) and A % B = lcm (A, B)
Evaluate the following postfix expression.
Note the numbers are single digits in the expression.

4 3 * 9 7 + % 4 5 + 2 ↑ 3 5 * @ /

17. 08-09 C4 Prefix/Infix/Postfix

Convert the following infix expression to prefix:

(A^2 + B) * C / B - (B + C) / A + A^3 * B^2

18. 08-09 C4 Prefix/Infix/Postfix

Define A @ B = gcf (A, B) and A % B = largest integer ≤ (A/B)
Evaluate the following postfix expression.
Note the numbers are single digits in the expression.

4 3 6 * 9 3 + @ 3 2 ↑ @ * 2 3 5 * * 7 5 * @ %

19. 09-10 C4 Prefix/Infix/Postfix

Convert the following postfix expression into prefix:

C A B ↑ B A ↑ + A B * ↑ * A B * C + B ↑ /

20. 09-10 C4 Prefix/Infix/Postfix

Define A @ B = gcf (A, B) and A % B = greatest integer ≤ (A/B)
Evaluate the following postfix expression. Note the numbers are single digits in the expression.

1 2 3 * + 3 2 ↑ + 2 2 ↑ 8 + @ 5 2 * 6 3 / - %

21. 10-11 C4 Prefix/Infix/Postfix

Convert the following postfix expression into prefix:

C A B ↑ B A ↑ + A B * ↑ * A B * C + B ↑ /

22. 10-11 C4 Prefix/Infix/Postfix

Define A @ B = gcf (A, B) and A % B = largest integer ≤ (A/B)
Evaluate the following postfix expression.
Note all the numbers are single digits..

4 3 6 * 9 3 + @ 3 2 ↑ @ * 2 3 5 * * 7 5 * @ %

23. 11-12 C2 Prefix/Infix/Postfix

Evaluate the following postfix expression. All numbers are single digits.

2 4 6 + * 5 / 3 2 ↑ 4 2 – * +

24. 11-12 C2 Prefix/Infix/Postfix

Convert the following prefix expression into a postfix expression.

+ – * A ↑ B C / B A * C + / A B C

25. 12-13 C2 Prefix/Infix/Postfix

Evaluate the following postfix expression if A = 2, B = 4, C = 6, D = 1, and E = 0:

A B + C / B E ↑ * D A + 2 ↑ + B C E * + A / –

26. 12-13 C2 Prefix/Infix/Postfix

Convert the following prefix expression into a postfix expression:

− + − * + A B −↑ A 2 B / + A B ↑ C 2 / A B C

27. 13-14 C2 Prefix/Infix/Postfix

Evaluate the following prefix expression if A = 8, B = 4, C = 2, and D = 6.

− + A B * / * C ↑ D C A B

28. 13-14 C2 Prefix/Infix/Postfix

Convert the following infix expression into a postfix expression.

$$\frac{A(B+C)}{A-C}+\frac{A+B}{C^2}$$

29. 14-15 C2 Prefix/Infix/Postfix

Convert the following infix expression to postfix:

A + B * (C + A^2) – B / (C + A)

30. 14-15 C2 Prefix/Infix/Postfix

Define a # b = (a+b)/2 (arithmetic mean) and a @ b = √ab (geometric mean). Evaluate the following prefix expression: (Note: all numbers are single digits)

− # * 8 6 / 4 2 − @ * @ 9 4 3 * ↑ 5 2 2 ↑ # 8 2 3

31. 15-16 C2 Prefix/Infix/Postfix

Convert the following infix expression to prefix:

$$\frac{(A+B)(C-A)(\frac{C}{B}-A)}{(AB-C)}$$

32. 15-16 C2 Prefix/Infix/Postfix

Evaluate the following postfix expression if a # b = $\frac{a+b}{2}$ and a % b = $\sqrt{ab}$:

6 4 # 9 4 % + 3 1 # ↑ 7 9 + 3 9 * 2 − % 5 / %